

Practical Assignments:

Assignment 1:

- 1) Write a C program to declare variables of type int, float, char, and double. Initialize them with values and print their values and sizes using sizeof operator.
- 2) Write a C program to input two integers and perform addition, subtraction, multiplication, division, and modulus. Display the results.
- 3) Write a C program to demonstrate the difference between ++x (pre-increment) and x++ (post-increment) in an arithmetic expression.

Assignment 2:

1. Problem Statement: Salary Calculation System

A company wants to automate its employee salary calculation process using a C program. The program should take the following inputs from the user:

- Employee name (string)
- Basic salary (float)

Using arithmetic operators, calculate and display the following:

1. House Rent Allowance (HRA) = 20% of basic salary
2. Dearness Allowance (DA) = 10% of basic salary
3. Gross Salary = Basic Salary + HRA + DA
4. Income Tax = 5% of Gross Salary
5. Net Salary = Gross Salary - Income Tax

Finally, display the employee name, basic salary, HRA, DA, Gross Salary, Income Tax, and Net Salary.

2. Problem Statement: Comparing Two Numbers Without Conditional Statements

A bank's internal tool needs to find out which of two account balances is higher, but the program cannot use any conditional statements (like **if**, **else**, or ternary operators).

Write a C program that:

- Takes two account balances as input (float).
- Uses relational and logical operators to determine:

- o Whether both balances are equal.
- o Whether the first balance is greater than the second.
- o Whether the second balance is greater than the first.
- Displays the results as 1 (True) or 0 (False) for each condition.

3. Problem Statement: Login Authentication System

A company wants to develop a simple login authentication system in C.

The system should check whether the entered username and password match the correct ones stored in the system.

Write a C program that:

- Stores a predefined username and password (e.g., "admin" and "1234").
- Takes username and password as input from the user.
- Uses logical operators (&&, ||, !) to check:
 1. If both username and password are correct, display “Login Successful.”
 2. If either username or password is incorrect, display “Invalid Login.”

4. Problem Statement: Bank Interest Update System

A bank wants to develop a simple program to update a customer’s account balance using assignment operators.

Write a C program that performs the following operations:

1. Takes the customer’s initial account balance as input.
2. Adds interest amount (using +=) to the balance.
3. Deducts annual maintenance charge (using -=) from the balance.
4. Multiplies the balance by a bonus factor (using *=) to reward loyal customers.

5. Divides the balance equally among two linked accounts (using /=).
6. Finally, displays the updated balance after each operation.

Assignment 3:

1. Write C program to print A to Z alphabets in star pattern
2. Write a C program to print your name in the star pattern.

Assignment 4:

1. Write a C program that uses a user-defined function to **swap two numbers using call by reference**. The function should take the addresses of the variables as parameters and modify their values.
2. Write a C program to **find the maximum of two numbers** using a function that accepts **pointer arguments**.